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CAPSTONE PROJECT

CHAPTER 5 - TESTING, RESULTS & EVALUATION

Student / Group Template

Project Title	SmartDesk: A Customer Support Ticketing System for Small IT Support Teams
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Module	SPM622 / MDB622
Lecturer / Supervisor	Mr. Dunstan
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5.1 Introduction

State the testing and evaluation purpose.

This chapter evaluates whether SmartDesk meets the requirements it was built against. It sets out the test strategy, reports test cases and results, records defects found and corrected, evaluates performance and usability, assesses the system against its requirements, summarises stakeholder feedback, and judges whether the system solves the identified problem.

5.2 Test Strategy

Explain test levels/types, environment, responsibilities and acceptance criteria.

Testing combined automated and manual methods.

Level	Method	Scope	Responsibility
Unit / integration	pytest with Flask test client	Auth, authorisation, ownership, ticket creation, comments	Wian
System	Manual, scripted	Full user journey across all three roles	Both
Security	Manual, adversarial	CSRF omission, cross-customer access, unauthorised role action, injection input	Wian
Usability	Observation	First-time ticket creation without instruction	Freek

The automated suite runs against a temporary database per test (`tmp_path`), so tests are isolated and repeatable. The three test functions are `test_register_create_and_comment`, `test_login_and_access_controls`, and `test_customer_isolation`.

Entry criteria: application starts, schema initialises, demo accounts seed.
Exit criteria: all automated tests pass, all Must-have requirements exercised manually, no open Critical or High defects.

5.3 Test Cases and Results

Reference the test-case workbook and summarise pass/fail results.

ID	Requirement	Test	Expected	Actual	Status
TC-01	Registration	Register with new email	Account created, dashboard shown	As expected	Pass

ID	Requirement	Test	Expected	Actual	Status
TC-02	Duplicate email	Register with existing email	Friendly error, no duplicate	As expected	Pass
TC-03	Weak password	Register with 4-char password	Rejected	As expected	Pass
TC-04	Login valid	Correct credentials	Session created, dashboard	As expected	Pass
TC-05	Login invalid	Wrong password	Identical error message	As expected	Pass
TC-06	Password storage	Inspect `users.password_hash`	Hash only, no plain text	As expected	Pass
TC-07	Logout	POST `/logout`	Session cleared, redirect to login	As expected	Pass
TC-08	Unauthenticated access	GET `/dashboard` without session	302 redirect to login	As expected	Pass
TC-09	CSRF omitted	POST without `_csrf`	HTTP 400	As expected	Pass
TC-10	Role restriction	Customer POSTs `/tickets/1/update`	HTTP 403	As expected	Pass
TC-11	Customer isolation	Customer 2 opens customer 1's ticket	HTTP 403	As expected	Pass
TC-12	Ticket creation	Valid title, description, category, priority	Ticket created, status Open	As expected	Pass
TC-13	Ticket validation	Empty title	Rejected, no insert	As expected	Pass
TC-14	Invalid enum	POST `priority=Urgent`	HTTP 400	As expected	Pass
TC-15	Agent assignment	Assign to agent	Assignee saved, status updated	As expected	Pass

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ID	Requirement	Test	Expected	Actual	Status
TC-16	Invalid assignee	Assign to a customer ID	HTTP 400	As expected	Pass
TC-17	Comment	Post comment on own ticket	Comment stored and displayed	As expected	Pass
TC-18	Empty comment	Post blank comment	Rejected	As expected	Pass
TC-19	Dashboard counts	Compare counts to ticket table	Counts reconcile	As expected	Pass
TC-20	Injection input	Title `'; DROP TABLE tickets;--`	Stored as literal text	As expected	Pass

5.4 Defects and Corrections

Describe important defects, their causes, fixes and retest outcomes.

ID	Feature	Description	Severity	Fix	Retest
D-01	Access control	Customer could open another customer's ticket by editing the URL	**Critical**	Ownership check moved into shared `get_ticket()` helper so every route inherits it	Pass (TC-11)
D-02	CSRF	POST endpoints accepted requests with no CSRF token	**High**	Central `before_request` guard using `secrets.compare_digest`	Pass (TC-09)
D-03	Assignment	Any user ID could be posted as assignee, including customers	**High**	Server-side SQL check that assignee holds `agent`/`admin` role	Pass (TC-16)
D-04	Validation	Free-text priority accepted on ticket creation	Medium	Allow-list validation against module-level tuples	Pass (TC-14)

ID	Feature	Description	Severity	Fix	Retest
D-05	Display	Long ticket titles broke dashboard table layout	Low	`maxlength` on input and CSS truncation	Pass (visual)

5.5 Performance / Usability Evaluation

Evaluate relevant non-functional requirements using evidence.

Measure	Method	Target	Result
Login response	Timed, 10 runs, local	≤ 2 s	*[0.4]* s
Dashboard load, 100 tickets	Timed, 10 runs	≤ 2 s	*[0.8]* s
Dashboard load, 1,000 tickets	Timed, 5 runs	≤ 3 s	*[1.9]* s
First-time ticket creation	Observed, unaided	≤ 60 s	*[42]* s
Validation message clarity	Review of all messages	Names field and fix	*[record]* of *[total]*

Usability observations. The dashboard leads with status counts, so backlog size is visible without scrolling. Priority is the only colour-coded element; status uses a neutral pill, which testers reported made triage faster. Login and registration share one layout, so users met the same form twice.

Limitation encountered. At 1,000 tickets the dashboard loads all rows before rendering rather than paging. This was measurable but did not breach the target on the test machine; it is recorded as future work.

5.6 Requirements Evaluation

Compare the completed system against functional and non-functional requirements.

Requirement area	Implemented	Evidence
Registration and login	Yes	TC-01, TC-04
Hashed credentials	Yes	TC-06

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Requirement area	Implemented	Evidence
Role-based access control	Yes	TC-10
Customer data isolation	Yes	TC-11, TC-12 (test)
CSRF protection	Yes	TC-09
Ticket creation with validation	Yes	TC-12, TC-13, TC-14
Ticket assignment to staff only	Yes	TC-15, TC-16
Status and priority management	Yes	TC-15
Attributable comments	Yes	TC-17
Status dashboard	Yes	TC-19
Parameterised SQL	Yes	TC-20, code review
Audit trail of changes	**No** — current state only, no history table	See 5.4 / Chapter 6
Enforced status transitions	**No** — any status settable	Chapter 6 limitation
Email notification, attachments, SLA timers	**No** — out of scope	Chapter 3 scope

Revision needed. The earlier SRS (FR-01 to FR-38) described a seven-table C# system with Technician/Administrator roles and enforced transitions. The delivered system differs. The SRS must be rewritten to match what exists before submission, or this evaluation cannot be reconciled with Chapter 3.

5.7 User / Stakeholder Feedback

Summarise feedback where applicable and explain resulting improvements.

Feedback came from five first-time users, two staff/administrator walkthrough participants and the supervisor's demo review; no external client was available, so the sample is indicative only. Three themes emerged: clarity of language, triage efficiency and workflow traceability. Clarity feedback led to clearer validation messages, priority helper text and a post-creation confirmation, while the dashboard's status counts and priority-only colour coding were retained. Audit history, enforced status transitions, notifications and unassigned-ticket filtering were valid but deferred to Chapter 6 as scope boundaries and future work.

5.8 Chapter Summary

Summarise whether the system adequately solves the identified problem.

SmartDesk was tested at unit, system, security and usability level. All three automated tests passed, twenty manual cases passed, and five defects were found and corrected — including one Critical (cross-customer access) and two High (CSRF, assignee validation). The system meets its core requirements: secure authentication, role-based authorisation, customer isolation, validated ticket creation, staff-only assignment, attributable conversation and a working dashboard. It does not implement an audit trail, enforced status transitions, notifications or attachments, and the dissertation must present these as scope boundaries and limitations rather than omissions. The system adequately solves the identified problem at MVP scale; Chapter 6 discusses the limitations and the work required to close them.

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